Day 39 - Feature Engineering and Association Rule Learning

1) Feature Engineering

Feature Engineering is one of the most important steps in Machine Learning.
It is the process of creating, selecting, transforming, and improving input features (columns) from raw data so that the machine learning model can perform better.
In simple words:
Definition
Feature Engineering is the process of converting raw data into useful features that help machine learning models learn better.

Example
Suppose we have a dataset with a column:
Date of Birth
From this one column, we can create new useful features like:

- Age
- Year
- Month
- Day

Weekday

This is called Feature Engineering.

Why Feature Engineering is Important

Feature Engineering is important because:

- It improves model accuracy
- It helps the model understand the data better
- It reduces irrelevant information
- It makes patterns easier to identify
- It improves training performance

2) What is Feature Engineering?

Feature Engineering means preparing better input variables for the machine learning model.
Machine learning models do not understand raw real-world data directly.
So we clean and transform the data into a useful format.

Simple Example

Suppose a dataset has:

Name	DOB	Salary
Ravi	10-03-2002	25000

From this we can create:

Age from DOB

Salary category from Salary

These newly created columns are called features.

3) Types of Feature Engineering

There are many types of feature engineering.

1. Feature Creation

Creating new features from existing data.

Example

From:

Date → create Year, Month, Day

Length and Width → create Area

Use

Helps model learn more meaningful patterns.

2. Feature Transformation

Changing the format or scale of data.

Examples

Log transformation

Square root transformation

Normalization

Standardization

Use

Helps when data is skewed or values are too large.

3. Feature Encoding

Converting categorical data into numerical form.

Machine learning models mostly work with numbers.

Examples

Male / Female → 0 / 1

Red / Blue / Green → One-Hot Encoding

Types

Label Encoding

One-Hot Encoding

4. Feature Scaling

Making all feature values come into the same range.

Why needed?

If one column has values in thousands and another has values in decimals, the model may become biased.

Methods

- Min-Max Scaling
- Standard Scaling

5. Feature Extraction

Extracting important information from existing data.

Example

From a text review:

- word count
- sentiment
- keywords

From images:

- edges
- shapes
- patterns

6. Feature Selection

Selecting only the most useful features and removing unnecessary ones.

Why?

Too many unnecessary columns can reduce performance.

Benefits

Reduces overfitting

Improves speed

Improves model accuracy

7. Handling Missing Values

Missing values are common in raw data.

Methods

Replace with mean

Replace with median

Replace with mode

Remove missing rows/columns

This is also part of feature engineering.

8. Binning / Discretization

Converting continuous values into groups.

Example

Age:

0-18 → Child

19-35 → Young

36-60 → Adult

This helps in pattern recognition.

4) What is Association Rule Learning?

Association Rule Learning is an unsupervised machine learning technique used to discover interesting relationships or patterns between items in a dataset.

It is mostly used in:

- Market basket analysis
- Product recommendation
- customer purchase behavior

Definition

Association Rule Learning is a rule-based machine learning method used to find relationships between items in large datasets.

Simple Example

Suppose customers buy:

- Bread
- Butter
- Milk

If many customers buy Bread, they may also buy Butter.

Then we can write the rule:

Bread→Butter

This means:

If a customer buys Bread, they are likely to buy Butter.

5) Types of Association Rule Learning

There are mainly two major algorithms used in association rule learning:

1. Apriori Algorithm

A classical algorithm used to find frequent itemsets and generate association rules.

2. ECLAT Algorithm

A faster algorithm that uses vertical data format and transaction IDs.

Sometimes association rules are also categorized as:

1. Boolean Association Rules

Items are either:

- present
- absent

Example:

- buys milk or not

2. Quantitative Association Rules

Used when values are numeric.

Example:

- age > 25
- income > 30000

6) Evaluation Metrics Used for Association Rule Learning

To check whether an association rule is useful or strong, we use some important metrics.

1. Support

Support tells how often an itemset appears in the dataset.

Formula

Support(A→B)=(Number" " of" " transactions" " containing" " A" " and" " B)/(Total" " number" " of" " transactions)

Meaning

Higher support means the rule appears more frequently.

2. Confidence

Confidence tells how often item B is bought when item A is bought.

Formula

Confidence(A→B)=(Support(A∪B))/(Support(A))

Meaning

Higher confidence means stronger rule.

3. Lift

Lift measures how strongly A and B are associated compared to random chance.

Formula

$$Lift(A \rightarrow B) = (Confidence(A \rightarrow B)) / (Support(B))$$

Interpretation

- Lift = 1 → No association
- Lift > 1 → Positive association
- Lift < 1 → Negative association

4. Leverage

Leverage tells the difference between actual co-occurrence and expected co-occurrence.

Meaning

It measures how much more often A and B occur together than expected.

5. Conviction

Conviction measures the dependency of the rule.

Meaning

Higher conviction means stronger implication.

Summary of Metrics

Metric	Meaning	Best Value
Support	Frequency of rule	High
Confidence	Reliability of rule	High
Lift	Strength of association	> 1
Leverage	Difference from expected	High
Conviction	Dependency of rule	High

7) What is Antecedent?

Antecedent means the "IF" part of an association rule.

Example

Bread→Butter

Here:

Antecedent = Bread

This is the item or condition that comes first.

8) What is consequent?

Consequent means the "THEN" part of an association rule.

Example

Bread→Butter

Here:

Consequent = Butter

This is the result or item likely to occur.

Simple Meaning

Antecedent = Left side of rule

Consequent = Right side of rule

9) Briefly Explain About the ECLAT Algorithm

ECLAT stands for:

Equivalence Class Clustering and Bottom-Up Lattice Traversal

It is an algorithm used in association rule learning to find frequent itemsets.

Main Idea of ECLAT

Instead of checking transactions one by one like Apriori, ECLAT uses:

Transaction ID Sets (TID Sets)

That means for each item, it stores the list of transaction IDs where that item appears.

Example

Suppose:

Transaction ID	Items
T1	Bread, Milk
T2	Bread, Butter
T3	Milk, Butter
T4	Bread, Milk, Butter

Then:

Bread → {T1, T2, T4}

Milk → {T1, T3, T4}

Butter → {T2, T3, T4}

Now to find support of Bread & Milk, ECLAT does intersection:

$$\{T1, T2, T4\} \cap \{T1, T3, T4\} = \{T1, T4\}$$

So support count = 2

Advantages of ECLAT

- Faster than Apriori in many cases
- Efficient for large datasets
- Uses intersection operation

Disadvantages

- Can use more memory
- Slightly difficult to understand compared

to Apriori

10) What is Apriori Algorithm? Explain the implementation

Apriori is one of the most popular algorithms used for Association Rule Learning.

It is used to find:

- Frequent itemsets
- Association rules

Main Idea of Apriori

Apriori is based on this principle:

"If an itemset is frequent, then all of its subsets must also be frequent."

This is called the:

Apriori Property

How Apriori Works

Apriori works in a step-by-step manner.

Step 1: Find Frequent 1-Itemsets

Find items that appear frequently.

Example:

- Bread

Milk

Butter

If they satisfy minimum support, keep them.

Step 2: Generate 2-Itemsets

Combine frequent 1-itemsets to form pairs.

Example:

Bread, Milk

Bread, Butter

Milk, Butter

Check support again.

Step 3: Generate 3-Itemsets

Combine frequent 2-itemsets to form larger sets.

Example:

Bread, Milk, Butter

Again check support.

Step 4: Remove Infrequent Itemsets

If support is below threshold, remove them.

Step 5: Generate Rules

From frequent itemsets, generate rules like:

Bread→Butter

Then calculate:

- Support
- Confidence
- Lift

Apriori Implementation (Simple Explanation)

Input

- Transaction dataset
- Minimum support
- Minimum confidence

Process

- Scan dataset
- Find frequent itemsets
- Remove low-support itemsets
- Generate rules
- Evaluate rules

Output

- Frequent itemsets
- Association rules

Simple Example

Suppose transactions are:

T1 → Bread, Milk

T2 → Bread, Butter

T3 → Milk, Butter

T4 → Bread, Milk, Butter

If minimum support is 2, then:

Frequent 1-itemsets:

Bread

Milk

Butter

Frequent 2-itemsets:

Bread, Milk

Bread, Butter

Milk, Butter

Frequent 3-itemset:

Bread, Milk, Butter

Then association rules are generated.

Advantages of Apriori

Easy to understand

Widely used

Good for market basket analysis

Disadvantages

Slow for very large datasets

Requires multiple scans of database

Generates many candidate itemsets

11) Use cases of Association Rule Learning

Association Rule Learning is widely used in real-world applications.

1. Market Basket Analysis

Used in supermarkets and online shopping.

Example:

If customer buys:

Bread

they may also buy:

Butter

2. Product Recommendation

E-commerce websites recommend related products.

Example:

If a user buys:

Mobile phone

they may also buy:

Mobile cover

Charger

3. Cross-Selling

Used by businesses to suggest related products.

Example:

Laptop → Mouse

4. Store Layout Optimization

Stores place related products near each other.

Example:

Chips and soft drinks together

5. Website Recommendation

Websites recommend related pages or items.

Example:

People who viewed one product also viewed another product

6. Healthcare

Used to identify relationships between:

symptoms

diseases

medicines

7. Banking and Finance

Used to detect spending or transaction patterns.

8. Fraud Detection
Helps find unusual or suspicious behavior patterns.

9. Telecom
Used to analyze customer usage and recharge patterns.

10. Education
Used to analyze student learning patterns and course preferences.